Summary of Revisions

Interpretable Multivariate Conformal Prediction with Fast Transductive Standardization

JMLR-25-3153-1

This summary compares the current draft with the original arXiv v1 submission of December 17, 2025. It describes changes present in the draft; red author checks identify unresolved decisions rather than completed changes.

- **Abstract.** The central contribution and finite-sample coverage claim are retained. Grammatical corrections improve readability; no new theoretical result is claimed.
- **Section 1: introduction and related work.** The motivation for interpretable simultaneous intervals and the Bonferroni comparison are retained. Figure 1 now shows illustrative marginal coverages of 92% and 93% alongside 90% joint coverage, making that distinction visible immediately. The final pass tightens the language. **AUTHOR-CHECK LIT:** The Point CHR/CQHR distinction, shape-template discussion, and empirical attribution still need substantive review.
- **Section 2: setup and motivating constructions.** The revised setup states joint exchangeability of calibration and test observations and explains that coverage is marginal over the data but simultaneous across outputs. Residual definitions discuss capping and ties. A common location/scale parameterization separates the population oracle from its practical approximations. These changes clarify the statistical target and motivation for coordinate adaptation; vector/scalar notation is standardized. **AUTHOR-CHECK MAP/TIES/BALANCE:** Review inverse terminology, tie handling, and the coverage-balance claim.
- **Section 3: methodology.** A threshold-indexed set map connects the transductive oracle, link functions, and global/local bounds. The local argument now gives a pointwise implication on oracle acceptance and working-region membership, not cell-conditional coverage. Figure 2 labels the residual boundaries; multi-indices are consistently bold. The real-data diagnostic reference now points to the appendix. These changes make the construction easier to follow. **AUTHOR-CHECK EVENT/ENDPOINT/BOUNDARY:** The piecewise proof, search-lemma endpoint, and partition boundaries still need mathematical review.
- **Section 4: numerical experiments. Twelve experimental studies have been added in total: ten simulation studies and two real-data diagnostic studies.** This count spans Section 4 and Appendix C and treats a parameter sweep as one study. Additional panels and coordinate plots are not counted separately. E11 and E12 share a rerun cohort, and E01 and E04 use the same dependent-Gaussian experiment family; the count is of distinct study questions, not independent datasets. The original independent Gaussian benchmarks, six-dataset comparison, and energy illustration remain.
- **E01: Dependent heterogeneous Gaussian noise (Section 4.2).** Tests whether coordinate adaptation remains useful when outputs are correlated. Joint coverage, volume, runtime, and coordinate lengths distinguish scale adaptation from the smaller but under-covering copula regions.
- **E02: Partial heteroskedasticity (Section 4.2).** Makes only five of ten coordinates input-dependent while preserving marginal second moments. Separates heteroskedasticity from unequal marginal scales and assesses joint, not input-conditional, coverage.
- **E03: CQR with native CQHR (Section 4.3).** Tests quantile-regression residuals and adds the requested CQHR baseline. Outcome-space volumes and lengths show design-dependent gains of capped TSCP while identifying its inability to shrink the fitted base interval.
- **Section 4.4: real-data presentation.** The original results are retained in an expanded table. A new plot of existing timing records makes wall-clock cost visible in the body; it is not counted as an added experiment. The Point CHR infinite volume on `stock` is explained by the small final calibration split. Coordinate and search diagnostics are presented in Appendix C.
- **Section 5: discussion.** The existing future directions are retained. New cross-references connect the outlier discussion to the contamination study and distinguish a direct signed-score extension from the capped and shifted CQR experiments.

- **Appendix A: algorithmic details and complexity.** The algorithms and complexity bounds are retained, with standardized vector notation and grammar. Branch diagnostics complement rather than improve the worst-case bound. **AUTHOR-CHECK DS:** Review the final-subset indexing in the data-splitting algorithm's quantile.
- **Appendix B: proofs.** The original arguments are retained with language, notation, punctuation, and display-layout corrections. No new theorem for infinite variance, ties, or signed residuals is claimed. **AUTHOR-CHECK DOMAIN/ZERO/ALGEBRA:** Review sample-size and degenerate cases, the zero-residual event argument, and the tagged quadratic coefficient.
- **Appendix C: supplementary experiments.** The original distribution tables, heavy-tail comparisons, population-oracle approximations, local-union comparison, and dimension-scaling study remain. The infinite-variance cases are explicitly outside the stated assumptions. The following nine added studies extend this evidence.
- **E04: Dependence-strength sweep (Appendix C.2).** Separates sensitivity to correlation from the single dependent setting in the body. Varying correlation at fixed marginal scales shows stable TSCP coverage and decreasing volume as dependence strengthens.
- **E05: Coordinate-heterogeneity sweep (Appendix C.3).** Identifies when coordinate rescaling is useful by varying the largest-to-smallest noise-scale ratio. The common unscaled threshold becomes less efficient as heterogeneity increases, while remaining competitive at equal scales.
- **E06: Target-coverage sweep (Appendix C.4).** Checks that the comparison is not specific to a 90 percent target. Targets of 80, 90, and 95 percent show the coverage-volume trade-off and the relative behavior of the methods.
- **E07: Exchangeable contamination (Appendix C.6).** Tests the sensitivity of mean and standard deviation scaling to outliers. Increasing row-wise contamination shows that near-nominal coverage can coexist with severe volume inflation.
- **E08: CQR base-interval sensitivity (Appendix C.7).** Checks dependence on the width of the fitted starting interval. The base-miscoverage sweep reveals CQHR's sensitivity to fitted widths and puts the main CQR comparison in context.
- **E09: Shifted CQR scores (Appendix C.7).** Examines additive shifts as an alternative to capping signed scores. Constants 100, 300, and 1000 give nearly identical recorded rectangles, equal to shifted TSCP-GWC within each run; this is not a general invariance theorem.
- **E10: Rectangular shape-template baseline (Appendix C.8).** Adds the requested comparison with the Tumu et al. family. The two-dimensional KDE/mean-shift study compares coverage, normalized volume, and runtime for a specific rectangular template implementation, not every multimodal design.
- **E11: Real-data coordinate diagnostics (Appendix C.10.1).** Goes beyond the original two-point energy illustration with all 51 target coordinates. Paired length ratios and marginal coverages show where volume gains do and do not shorten individual intervals, including unfavorable student-data results.
- **E12: Real-data search and fallback diagnostics (Appendix C.10.2).** Measures actual branch use, inspected candidates, and construction times. No backward search or fallback occurs at miscoverage 0.1 in the tested splits; larger miscoverage levels exercise both branches without demonstrating a worst-case scan.
- **Figures, references, and final checks.** The current draft uses 17 added experimental PDF assets and eight restyled original experimental assets; several assets can belong to one study. The Tumu et al. reference and both schematic annotations are retained. No numerical data or figure PDF was changed in the final copyedit. **AUTHOR-CHECK DATA/SHIFT/REPRO/COLOR/METADATA:** Resolve sampling provenance, the shift explanation/selection rule, run documentation, revision highlighting, and template metadata before submission.